

Introductory Statistics for Econometrics

Unit 2. Random Variables and Univariate Distributions

Bayes' theorem, random variables,
probability mass and density functions, the c.d.f.

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Learning outcomes. After this unit, the student should be able to (i) apply Bayes' theorem to update probabilities; (ii) define a random variable as a measurable map from the sample space to $\mathbb{R}$; (iii) distinguish discrete, continuous and mixed random variables; (iv) compute and interpret probability mass functions, probability density functions and cumulative distribution functions; (v) recognise the Bernoulli, Binomial and uniform distributions.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 4–6.

1 Bayes' theorem

The Law of Total Probability decomposes $\mathbb{P}(A)$ in terms of conditional probabilities $\mathbb{P}(A | B_i)$. Bayes' theorem reverses the direction of conditioning and is the natural tool whenever we update beliefs in light of data.

Theorem 1.1 (Bayes' theorem). *Let $\{A_1, A_2, \dots\}$ be a partition of the sample space $\mathcal{S}$ with $\mathbb{P}(A_i) > 0$ for every i , and let B be an event with $\mathbb{P}(B) > 0$. Then*

$$\mathbb{P}(A_i | B) = \frac{\mathbb{P}(B | A_i) \mathbb{P}(A_i)}{\sum_{j \geq 1} \mathbb{P}(B | A_j) \mathbb{P}(A_j)}. \quad (1)$$

Proof. By the multiplication rule, $\mathbb{P}(A_i \cap B) = \mathbb{P}(B | A_i) \mathbb{P}(A_i)$. Dividing by $\mathbb{P}(B)$ and using the Law of Total Probability for the denominator gives (1). $\square$

The terminology of *prior* and *posterior* probabilities will reappear throughout statistics:

- $\mathbb{P}(A_i)$: the *prior* probability of state A_i — what we believe *before* observing B .
- $\mathbb{P}(A_i | B)$: the *posterior* probability — what we believe *after* observing B .
- $\mathbb{P}(B | A_i)$: the *likelihood* of the data B under state A_i .

Example 1.2 (Diagnostic test). A test for a disease has sensitivity $\mathbb{P}(B | A) = 0.99$ (probability that the test is positive when the patient has the disease) and false-positive rate $\mathbb{P}(B | A^c) = 0.05$. The disease is rare: $\mathbb{P}(A) = 0.005$. Given a positive test, what is the probability that the patient is in fact sick?

$$\mathbb{P}(A | B) = \frac{0.99 \cdot 0.005}{0.99 \cdot 0.005 + 0.05 \cdot 0.995} \approx 0.090.$$

Despite the test's high sensitivity, the posterior probability of disease is under 10% because the disease is rare. The cognitive failure of ignoring the prior is known as *base-rate neglect*.

Example 1.3 (Survivorship bias). Suppose all surviving sailors of shipwrecks (event A) report having prayed (event B): $\mathbb{P}(B \mid A) = 1$. If non-survivors also pray with the same probability, $\mathbb{P}(B \mid A^c) = 1$, then by (1) $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ — praying is uninformative. Drawing causal conclusions from observing only the survivors is the simplest example of *selection bias*, a key concern in applied econometrics.

Remark 1.4 (Sequential updating). Bayesian updating is *coherent*: the posterior obtained after processing data B then C separately equals the posterior obtained from $B \cap C$ jointly. This will be the structural property behind the likelihood principle in Unit 9.

2 Random variables

We rarely care about the raw outcome $s \in \mathcal{S}$; we care about a numerical *summary* — number of heads, returns on a portfolio, hours unemployed. A random variable is exactly such a summary.

Definition 2.1 (Random variable). A (real-valued) *random variable* is a function $X : \mathcal{S} \rightarrow \mathbb{R}$ that maps each outcome $s \in \mathcal{S}$ to a real number $X(s)$.

The probability $\mathbb{P}$ on $\mathcal{S}$ induces a probability distribution for X : for any $A \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A) \equiv \mathbb{P}(\{s \in \mathcal{S} : X(s) \in A\}). \quad (2)$$

In practice we work directly with $\mathbb{P}(X \in A)$ and forget the underlying $\mathcal{S}$.

Remark 2.2. A random variable that takes only the values 0 and 1 is called an *indicator*. We write $\mathbf{1}_A = 1$ if A occurs and $\mathbf{1}_A = 0$ otherwise. Indicators reduce probabilities to expectations: $\mathbb{P}(A) = \mathbb{E}[\mathbf{1}_A]$.

We classify random variables according to the size of their range:

- *Discrete*: X takes at most countably many values.
- *Continuous*: X takes values in an interval of $\mathbb{R}$ and the probability of any particular point is zero.
- *Mixed*: a combination of the two — common in censored data (e.g. daily rainfall, top-coded incomes).

3 Discrete random variables and the p.m.f.

Definition 3.1 (Probability mass function). If X is discrete with possible values $\{x_1, x_2, \dots\}$, its *probability mass function* (p.m.f.) is

$$f_X(x) = \mathbb{P}(X = x), \quad x \in \mathbb{R}.$$

A p.m.f. satisfies $f_X(x) \geq 0$ and $\sum_i f_X(x_i) = 1$. For any $A \subseteq \mathbb{R}$, $\mathbb{P}(X \in A) = \sum_{x_i \in A} f_X(x_i)$.

Discrete uniform. X is *discrete uniform* on $\{x_1, \dots, x_K\}$ if $f_X(x_i) = 1/K$ for $i = 1, \dots, K$.

Bernoulli. $X \sim \text{Bernoulli}(p)$ if $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. Bernoulli is the workhorse for binary outcomes — employment, default, treatment status — and the building block for the binomial distribution.

Binomial. Suppose we run n independent Bernoulli(p) trials and let X be the total number of successes. Then

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n, \quad (3)$$

written $X \sim B(n, p)$. The binomial coefficient counts the $\binom{n}{x}$ orderings, while $p^x(1-p)^{n-x}$ is the probability of *any* particular ordering with x successes.

Example 3.2 (Bayesian updating with the binomial). You have a coin that may be either fair ($p_F = 1/2$) or biased ($p_L = 4/5$); your prior on “biased” is $\mathbb{P}(A) = 1/9$. You toss the coin 10 times and observe 8 heads (event B). The likelihoods are $\mathbb{P}(B | A) = \binom{10}{8} (4/5)^8 (1/5)^2$ and $\mathbb{P}(B | A^c) = \binom{10}{8} (1/2)^{10}$. Bayes’ theorem gives $\mathbb{P}(A | B) \approx 0.46$. The data raised the posterior probability that the coin is biased from 11% to 46%, but a single experiment is not yet conclusive — repeating the experiment refines the posterior further.

4 Continuous random variables and the p.d.f.

When X takes values in a continuum, the probability of any single point is zero, so we cannot describe its distribution by point probabilities. We describe it instead by a *density*.

Definition 4.1 (Probability density function). A continuous random variable X has *probability density function* (p.d.f.) $f_X : \mathbb{R} \rightarrow [0, \infty)$ if for every interval $A \subseteq \mathbb{R}$,

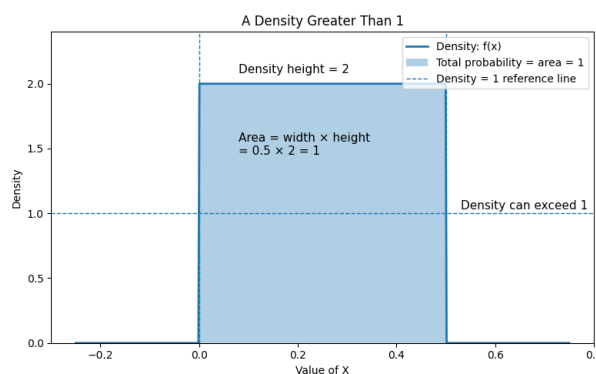
$$\mathbb{P}(X \in A) = \int_A f_X(t) dt.$$

A p.d.f. must satisfy $f_X(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f_X(t) dt = 1$.

Remark 4.2. $f_X(x)$ is *not* a probability — it can exceed 1. Probability is obtained by integrating f_X over a region. Consequently $\mathbb{P}(X = x) = 0$ for every $x \in \mathbb{R}$ when X is continuous, so $\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X \leq b)$. Densities can be interpreted as the frequency of an outcome.

Continuous uniform. $X \sim U[a, b]$ if

$$f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x), \quad F_X(x) = \frac{x-a}{b-a} \text{ on } [a, b].$$



A density function is not itself a probability. For example, if

$$X \sim U[0, 0.5],$$

then its density is

$$f_X(x) = \frac{1}{0.5 - 0} = 2 \quad \text{for } x \in [0, 0.5].$$

Thus, the density takes the value $2 > 1$. This is not a contradiction, because probabilities are given by areas under the density curve, not by the height of the curve itself. In this case,

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_0^{0.5} 2 dx = 2(0.5) = 1.$$

Therefore, a density may be greater than one at some points, as long as its total area equals one.

Example 4.3. Let $f_X(x) = ax^2$ on $(0, 3)$ and zero otherwise. The constant a is fixed by the normalisation $\int_0^3 ax^2 dx = 9a = 1$, so $a = 1/9$. Then

$$\mathbb{P}(1 < X < 2) = \int_1^2 \frac{t^2}{9} dt = \frac{2^3 - 1^3}{27} = \frac{7}{27}.$$

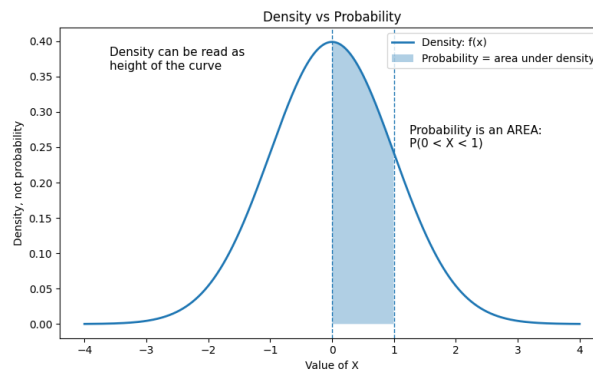
Example 4.4. The usual Gaussian bell curve represents the **density** of the normal distribution. The **area under the density function between two points** gives the probability that the random variable falls between those two values. Now suppose we shrink the interval around a point, for example around 0. As the two endpoints get closer and closer together, the area under the curve goes to zero. Therefore,

$$\mathbb{P}(X = 0) = 0.$$

However, the density at that point is still positive:

$$f(0) > 0.$$

This illustrates an important feature of continuous random variables: the probability of taking any exact value is zero. Probabilities are assigned to intervals, not to individual points.



5 The cumulative distribution function

The c.d.f. provides a single object that handles discrete, continuous and mixed random variables uniformly.

Definition 5.1 (Cumulative distribution function). The *cumulative distribution function* (c.d.f.) of X is

$$F_X(x) = \mathbb{P}(X \leq x), \quad x \in \mathbb{R}.$$

Proposition 5.2 (Properties of a c.d.f.). Any c.d.f. F_X satisfies:

- (i) $0 \leq F_X(x) \leq 1$ for every x ;
- (ii) F_X is non-decreasing;
- (iii) $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$;

(iv) F_X is right-continuous: $F_X(x) = F_X(x^+)$.

Conversely, any function with these four properties is the c.d.f. of some random variable.

Proposition 5.3 (Probabilities from the c.d.f.). For any $a < b$ and any $x \in \mathbb{R}$,

$$\begin{aligned}\mathbb{P}(X > x) &= 1 - F_X(x), \\ \mathbb{P}(a < X \leq b) &= F_X(b) - F_X(a), \\ \mathbb{P}(X < x) &= F_X(x^-) := \lim_{h \downarrow 0} F_X(x - h), \\ \mathbb{P}(X = x) &= F_X(x^+) - F_X(x^-).\end{aligned}$$

The last identity is the key reason to define the c.d.f. as $\mathbb{P}(X \leq x)$ rather than $\mathbb{P}(X < x)$: jumps of F_X correspond exactly to point masses, and continuous distributions are recognised as the ones whose c.d.f. has no jumps.

Relation between p.d.f./p.m.f. and c.d.f.

- Discrete: $F_X(x) = \sum_{x_i \leq x} f_X(x_i)$ (a step function).
- Continuous: $F_X(x) = \int_{-\infty}^x f_X(t) dt$, hence by the Fundamental Theorem of Calculus $f_X(x) = F'_X(x)$ wherever the derivative exists.

Example 5.4 (Discrete c.d.f.). For a fair die, $X \in \{1, \dots, 6\}$ with $f_X(k) = 1/6$. The c.d.f. is the step function

$$F_X(x) = \begin{cases} 0, & x < 1, \\ k/6, & k \leq x < k+1, \quad k = 1, \dots, 5, \\ 1, & x \geq 6. \end{cases}$$

Example 5.5 (Continuous c.d.f.). Continuing Example 4.3, $f_X(x) = x^2/9$ on $(0, 3)$ gives $F_X(x) = x^3/27$ on the same interval, with $F_X(x) = 0$ for $x \leq 0$ and $F_X(x) = 1$ for $x \geq 3$. Conversely, differentiating F_X on $(0, 3)$ recovers the density: $F'_X(x) = 3x^2/27 = x^2/9 = f_X(x)$.

6 Mixed distributions

Many economic variables are neither purely discrete nor purely continuous. For example, daily rainfall is zero with positive probability and otherwise positive-real-valued; reported income often shows a point mass at zero (non-employment) or at upper limits of tax brackets. The corresponding c.d.f. has *jumps* (point masses) and *slopes* (densities) simultaneously. The general decomposition is

$$F_X(x) = \alpha F^{\text{disc}}(x) + (1 - \alpha) F^{\text{cont}}(x), \quad \alpha \in [0, 1].$$

We will return to mixed distributions when modelling censored and corner-solution outcomes in econometrics.

Summary

- *Bayes' theorem.* $\mathbb{P}(A_i | B) = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B|A_j)\mathbb{P}(A_j)}$. Reverses the direction of conditioning; turns prior + likelihood into posterior.
- *Random variable.* A function $X : \mathcal{S} \rightarrow \mathbb{R}$. Its distribution is induced by $\mathbb{P}$.
- *Discrete:* p.m.f. $f_X(x) = \mathbb{P}(X = x)$. Examples: discrete uniform, Bernoulli, Binomial $B(n, p)$.
- *Continuous:* p.d.f. $f_X \geq 0$ with $\int f_X = 1$; $\mathbb{P}(X \in A) = \int_A f_X$.
- *c.d.f.:* $F_X(x) = \mathbb{P}(X \leq x)$ — non-decreasing, right-continuous, with limits 0 and 1. Discrete c.d.f. is a step function; continuous c.d.f. is differentiable with $F'_X = f_X$.

Suggested exercises

1. Two boxes. Box 1 contains 3 red and 2 blue balls; Box 2 contains 1 red and 4 blue. A box is chosen uniformly at random and a ball drawn from it; the ball is red. What is the posterior probability that Box 1 was chosen?
2. Let $X \sim B(10, 0.3)$. Compute $\mathbb{P}(X = 3)$, $\mathbb{P}(X \leq 2)$ and $\mathbb{P}(X \geq 8)$.
3. Verify that $f(x) = 2x$ on $(0, 1)$ is a valid p.d.f. Compute the c.d.f., $\mathbb{P}(X < 1/2)$, $\mathbb{P}(1/4 < X < 3/4)$ and $\mathbb{P}(X = 1/2)$.
4. Let $F(x) = 0$ for $x < 0$, $F(x) = x/(1+x)$ for $x \geq 0$. Verify that F is a c.d.f., compute its p.d.f., and find $\mathbb{P}(X > 2)$.
5. (Monty Hall.) Three doors A, B, C . Behind one is a prize. You pick A ; the host (who knows where the prize is) opens C , revealing no prize. Use Bayes' theorem to compute $\mathbb{P}(\text{prize behind } B \mid C \text{ opened})$ and conclude that switching doors raises the winning probability from $1/3$ to $2/3$.